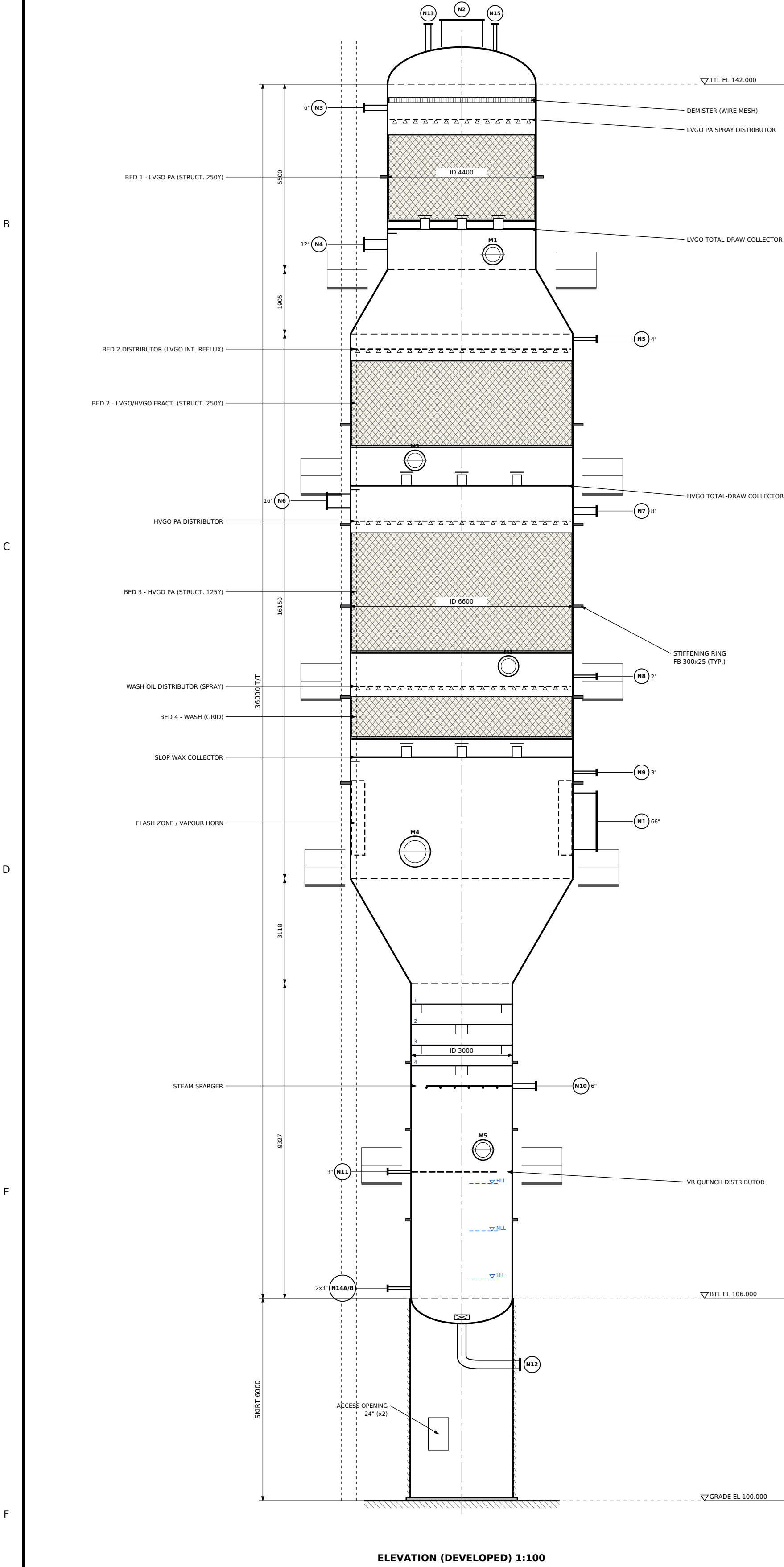




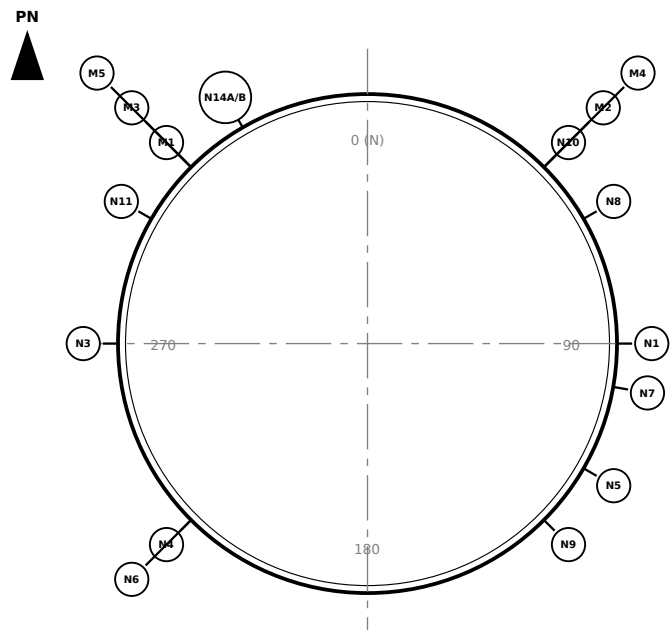
NOZZLE SCHEDULE						
MARK	QTY	SIZE	RATING	LOCATION	ORIENT	SERVICE
N1	1	66"	150# RF	EL 120.145	90	Feed - transfer line from H-201 (tangential to vapour horn)
N2	1	48"	150# RF	TOP HEAD	0	Overhead vapour to J-201 / E-202
N3	1	6"	150# RF	EL 141.300	270	LVGO PA return (x LVGO reflux)
N4	1	12"	150# RF	EL 137.250	225	LVGO total draw to P-201
N5	1	4"	150# RF	EL 134.445	120	LVGO internal reflux to bed 2
N6	1	16"	150# RF	EL 129.645	225	HVGO total draw to P-202
N7	1	8"	150# RF	EL 129.345	100	HVGO PA return
N8	1	2"	150# RF	EL 124.445	60	Wash oil (HVGO) to bed 4
N9	1	3"	150# RF	EL 121.595	135	Slop wax draw to P-203
N10	1	6"	150# RF	EL 112.297	45	Stripping steam
N11	1	3"	150# RF	EL 109.750	300	VR quench return (from E-201 outlet)
N12	1	10"	150# RF	BTM HEAD	0	Vacuum residue to P-204
N13	1	6"	150# RF	TOP HEAD	0	PSV-2001 relief (R)
N14A/B	2	3"	150# RF	EL 106.300	330	LT / LG bridle (boot)
N15	1	4"	150# RF	TOP HEAD	0	Vent / steam-out / N2 purge
M1	1	24"	150# RF	EL 136.950	0	Manway 24"
M2	1	24"	150# RF	EL 130.845	0	Manway 24"
M3	1	24"	150# RF	EL 124.745	0	Manway 24"
M4	1	36"	150# RF	EL 119.245	0	Manway 24" (36" in flash zone)
M5	1	24"	150# RF	EL 110.400	0	Manway 24"

HYDRAULIC SECTIONS (PROCESS SIZING)						
SECTION	V t/h	L t/h	rho V	u allow/ft	D calc	D sel.
Bed 1 - LVGO PA	53	83	0.110	9.39	4.27	4.40
Bed 2 - LVGO/HVGO fract.	87	34	0.176	7.11	4.97	6.60
Bed 3 - HVGO PA	178	161	0.283	5.39	6.41	6.60
Bed 4 - Wash	152	13	0.307	5.77	5.50	6.60
Flash zone	152	108	0.327	5.23	5.60	6.60
Stripping	5	108	0.041	11.12	2.49	3.00

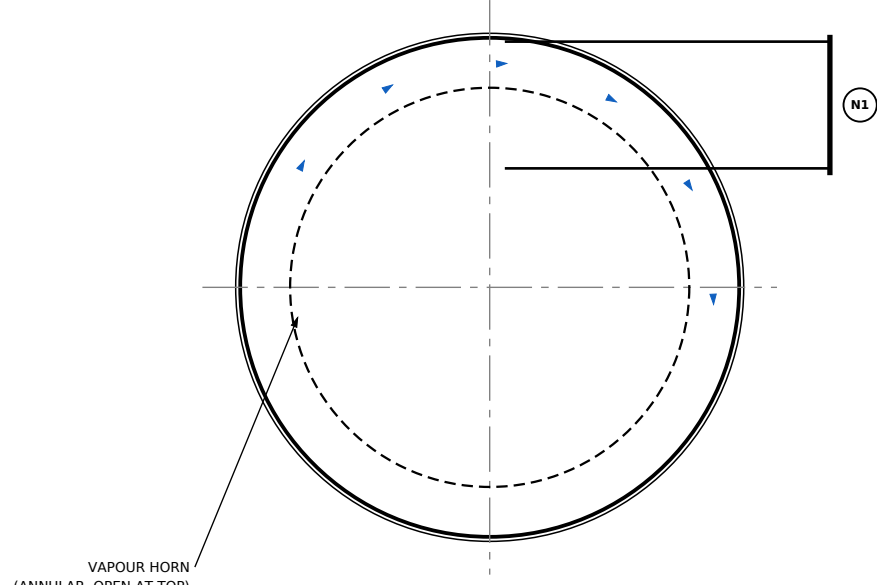
EXTERNAL PRESSURE UG-28/29 - FULL VACUUM (1.034 bar)						
COURSE	t nom	t ext	L / Le mm	Pa bar	RINGS	FB h x b
Boot (stripping / quench)	28	18	3000	1.222	3	160x16
Lower cone 30 deg	25	25	2267	1.141	-	-
Main shell (beds 2-4, flash zone)	25	24	3000	1.070	5	300x25
Upper cone 30 deg	22	22	1588	1.079	-	-
Top section (bed 1)	22	21	3000	1.226	1	220x20

SHELL COURSE / THICKNESS SCHEDULE						
COMPONENT	ID mm	FROM EL	TO EL	t mm	MATERIAL	REMARK
Skirt	3056 OD	100.000	106.000	20	SA-516-70	fireproofed
Bottom head 2:1	3000	-	106.000	18	SA-516-70	
Boot (stripping / quench)	3000	106.000	115.327	28	SA-516-70	t req 12.2 / ext 18
Lower cone 30 deg	6600 cone	115.327	118.445	25	SA-516-70	t req 20.7 / ext 25
Main shell (beds 2-4, flas	6600	118.445	134.595	25	SA-516-70	t req 18.8 / ext 24
Upper cone 30 deg	6600 cone	134.595	136.500	22	SA-516-70	t req 20.7 / ext 22
Top section (bed 1)	4400	136.500	142.000	22	SA-516-70	t req 14.5 / ext 21
Top head 2:1	4400	142.000	-	22	SA-516-70	

PLATFORM SCHEDULE			
PLATF.	ELEVATION	SERVES MANWAY	NOZZLES ACCESSED
P1	EL 109.400	M5	N11
P2	EL 118.240	M4	N1
P3	EL 123.750	M3	N8
P4	EL 129.840	M2	N6, N7
P5	EL 135.950	M1	N4
P6	EL 143.300	top head / davit	

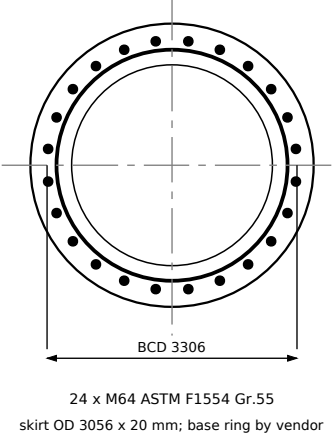


NOZZLE ORIENTATION PLAN (MAIN SHELL) 1:100
(manways alternate 315 / 45 deg - platform side)



SECTION THROUGH FLASH ZONE 1:100
tangential feed, 270 deg horn

SKIRT BASE / ANCHOR BOLT PLAN 1:100



DESIGN DATA	
Service	Vacuum column (wet, packed)
Design pressure	3.5 barg / FULL VACUUM
Design temperature	415 C
Operating P / T	20 / 60 mbar(a) / 70 / 386 C
Shell material	SA-516-70
Cladding / lining	CS + 4105 clad (317L clad flash/wash zone)
Corrosion allowance	6.0 mm
Joint efficiency	1.0 (full RT)
Shell thickness	28, 25, 25, 22, 22 mm (by course, bottom up)
Heads (2:1 SE) top / btm	22 / 18 mm
Skirt OD x t x height	3056 x 20 x 6000 mm
Anchor bolts	24 x M64 on 3306 BCD
Hydrotest (top / btm)	6.91 / 10.62 barg
Insulation	125 mm mineral wool
Weight empty / oper.	320 / 367 t
Weight hydrotest	1128 t
Wind base V / M	692 km / 17661 kNm
Code	ASME VIII Div.1, U-stamp; ASCE 7-16

LEGEND	
	Valve tray with downcomer
	Liquid distributor
	Liquid level (LLL/LL/LLL)
	Nozzle - mark and size
	Draw sump
	Packed bed (structured / grid)
	Collector / chimney tray
	Manway (shown on face)
	Platform with handrail
	External stiffening ring

WEIGHTS & LOADS	
Shell + heads + nozzles + clad	148.0 t
Skirt + base ring	11.7 t
Internals	107.4 t
Platforms / ladders	26.4 t
Insulation + fireproofing	26.9 t
EMPTY (installed)	320.5 t
OPERATING	366.9 t
HYDROTEST (full of water)	1128.0 t
Wind base shear / moment (ASCE 7, strength)	692 kN / 17661 kNm
Natural frequency / gust factor	0.74 Hz / 1.22

- NOTES**
- Dimensions in mm, elevations in m. Grade = EL 100.000.
 - Thicknesses are FEED minimum nominal (CFU-000-ME-CAL-001); fabricator to verify.
 - Nozzle orientations preliminary - to be confirmed by piping study.
 - Platforms 1200 wide, 60 % wrap; caged ladders not shown in full.
 - Datasheet CFU-000-ME-DS-001; weights in data/mech.json.
 - Design full vacuum: external stiffening rings at max. 3.0 m spacing (see table).
 - Boot: VR quench distributor above HLL limits boot residence/coking.
 - HOLD: top section ID per equipment.json - see CAL-001 S13 (bed 1 rating).

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT		
COLUMN GENERAL ARRANGEMENT C-201 VACUUM COLUMN		
DISCIPLINE	STAGE	SCALE
MECHANICAL	FEED	1:100
DRAWN	CHECKED	APPROVED
CLAUDE CODE	-	-
DRAWING NUMBER	CFU-200-ME-GA-004	
DATE	2026-10-02	REV
1 OF 1		A